

Andrew Desmedt

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EXPERIENCE

STEL Controls - Fredericksburg, VA — *Junior Mechatronics Engineer*

August 2025 - Present

- Developed Python control software for a precision mechatronic system achieving sub-millimeter repeatability; integrated motion planning and external subsystem communication.
- Support the system's integration with FANUC robot, Rockwell PLC, and AI-based vision.
- Conducted prototyping, testing, and troubleshooting.
- Document code, processes, and system configurations.
- Designed and prototype a 3 nozzle paint-jet head.
- CAM and CNC with 3d toolpaths.
- Program microcontrollers and develop circuits with over the air updates.

Team 1731 Fresta Valley Robotics — Captain, Mentor

Captain of robotics team - 2025; Mentor - 2026

- Led a competitive robotics team, coordinating mechanical, electrical, and software subsystems.
- Experience working with electrical, mechanical, and software systems
- Won two district events and qualified for world championship
- Mentored younger team members.
- Came up with a solution that enabled the robot to complete an extremely valuable task.

Andy's Heritage Hens - Nokesville, VA — *Owner/Manager*

April 2019 - May 2023



- Managed care of up to 70 laying hens.
- Delivered fresh eggs to 20 customers weekly
- Managed all aspects of the business, including budgeting, marketing, customer service, and logistics every day for 4 years.
- Built a custom chicken coop with an automatic coop door to improve efficiency. This was my first experience with robotics.

SKILLS

Java, Python, C++

Microcontrollers: ESP32, STM32
Arduino

CAD/CAM - Onshape, Fusion,
Inventor, Autocad

Github

CNC Machining

Robotics

Microsoft Office

Troubleshooting

3d Printing: FDM/Resin

Customer service

Sales & Marketing

Project Management

AWARDS

2025 Innovation in control Award First Chesapeake Glenn Allen VA Event

2025 Winner First Chesapeake Glenn Allen VA Event

2025 Winner First Chesapeake Bethesda MD Event

2026 Awardee Outstanding Achievement in Engineering and Engineering Technology, Laurel Ridge Community College

LANGUAGES

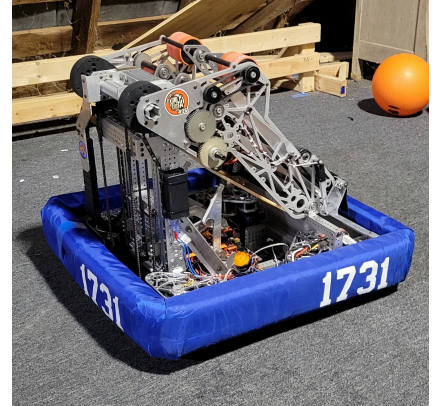
Duolingo Chinese Score 28

Duolingo Russian Score 28



Certificates

ETA International Fiber Optic
Installer



2024 Crescendo Robot “LowRider”

EDUCATION

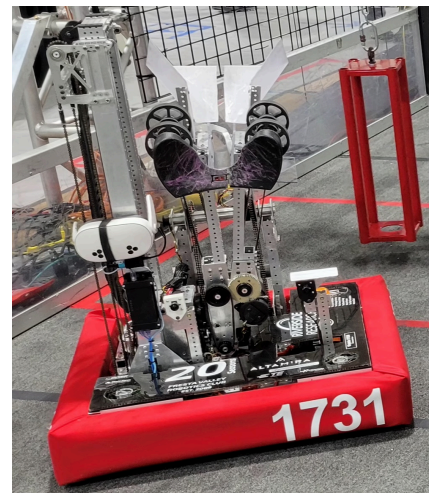
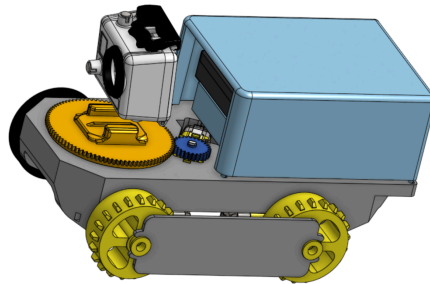
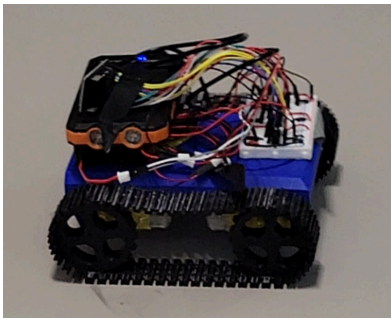
Virginia Polytechnic Institute and State University —
Bachelor of Applied Science in Electrical Engineering

July 2026 - Present

Laurel Ridge Community College, Warrenton — Associate of
Applied Science in Engineering

September 2023 - May 2026

- Designed, wired, and programmed a remote control vehicle.
- Coursework includes mechanical systems, electronics, and CAD.



2025 Reefscape Robot “MacDyver”

Phi Theta Kappa Honor Society — Member

Epsilon Pi Tau Honor Society — Member